AgriScanAI

Plant Foliar Disease Diagnostic System

Artificial Intelligence for Sustainable Agriculture.

AgriScan AI uses Artificial Intelligence (AI) to identify plant diseases from leaf images. It helps farmers detect diseases faster and make better farming decisions.

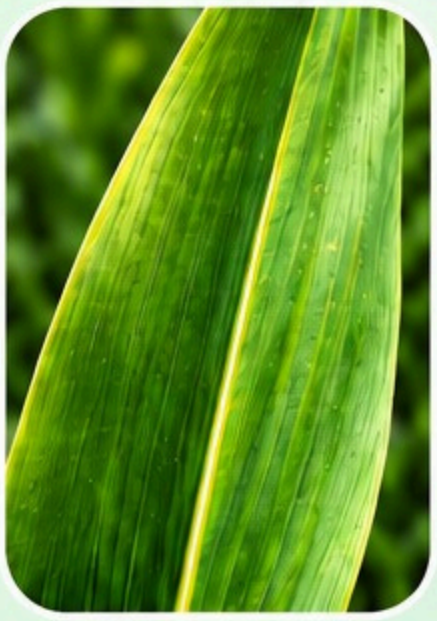
PROBLEMS

- Manual diagnosis is slow & expensive
- Excessive pesticide use harm the environment.
- Farmers lack access to experts
- Late diagnosis can reduce crop yields.

SOLUTIONS

- CNN-based automated disease detection
- Real-time prediction from leaf images
- 38 disease types across 13 crops
- Instant results with confidence scores.

PREDICTION



Plant:
Corn(Maize)

Disease:
Healthy Corn Leaf

Status:
HEALTHY

Confidence:
100.00%

HOW IT WORKS

- **Input:** Plant leaf image.
- CNN model analyzes patterns.
- **Output:** Disease class + confidence.
- Provides symptoms, prevention, treatment.

RESULTS

- Accuracy: ≥85%
- Prediction time: <3 seconds.
- Confident & reliable diagnoses.

DATA

- Plant Village dataset: 54,000+ images.
- 38 disease classes.
- 13 crop types.
- Preprocessed & augmented.

KEY FEATURES

- Instant diagnosis.
- Top 3 predictions.
- Disease knowledge base.
- CLI & API access.

MODEL

- 4-layer CNN architecture.
- Input: 128x128 RGB images.
- ~2.5M parameters.
- Trained with class weighting.

IMPACT

- Reduces costs & increases yields.
- Democratizes expert knowledge.
- Reduces pesticide use.
- Sustainable farming.